

The Physics Loss Channel: Consistency Is Not Accuracy

Sehaj Randhir Singh

*Department of Electrical and Computer Engineering,
NYU Tandon School of Engineering (independent researcher)
Brooklyn, NY, USA
sehajrsinghs@gmail.com*

TABLE I
LOSS-CHANNEL MATRIX: MEAN HELD-OUT REL-MAE (5 SEEDS)
WITH PHYSICS IN THE LOSS (PHYS), WITHOUT (NOPHYS), AND
WITH AN MLP HEAD; PER-DOMAIN PAIRED p AND EFFECT SIZE δ .

Domain	phys	nophys	MLP	p	δ
beam	0.147	0.193	0.156	0.438	-0.20
cantilever	0.148	0.204	0.152	0.188	-0.60
projectile	0.108	0.076	0.065	0.125	+0.60
burgers	0.009	0.010	0.018	0.438	-0.20
heat2d	0.015	0.014	0.016	0.625	+0.60

Abstract—Adding a physics residual to the loss is the standard way to make a network physical – and the standard way to stop measuring what it does. We isolate the loss channel on a controlled matrix: identical architectures, data, and training, with the physics term on or off. The physics term cuts the governing-equation residual $19\times$ ($p \leq 1e-7$, Cliff’s $\delta = -0.92$) – it genuinely enforces consistency – while pooled held-out accuracy stays flat ($p = 0.65$) and one domain (projectile) gets worse. Consistency is not accuracy; the loss channel is real, separable, and insufficient. The full 75-run matrix (5 domains $\times$ 3 architectures $\times$ 5 seeds) is committed and re-runnable.

Index Terms—physics-informed neural networks; loss design; PINN; verification

I. INTRODUCTION

PINN-style losses are ubiquitous and rarely controlled. This study separates two effects that a single number usually conflates: does the physics term make predictions more consistent with the governing equation, and does it make them more accurate?

II. PROTOCOL

Three model kinds (physics-in-the-loss transformer, no-physics transformer, MLP head) $\times$ five domains $\times$ five seeds, pooled paired statistics (Wilcoxon signed-rank with exact permutation, Cliff’s delta). Data, splits, and training are identical across kinds.

III. RESULTS

The pooled residual reduction is $19\times$ ($p \leq 1e-7$); pooled accuracy is null ($p = 0.65$) and the direction varies

by domain. The companion physics-transformers paper shows the channel that does move accuracy: physics as input tokens, causally active at inference.

REFERENCES

- [1] M. Raissi, P. Perdikaris, and G. E. Karniadakis, “Physics-informed neural networks,” *Journal of Computational Physics*, vol. 378, pp. 686–707, 2019.